

Assignment 2 Defense Notes - ID3

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Quick Setup

- Classifier Choice: ID3
- Live Terminal Output: Yes
- T1 Threshold: 0.8
- g Parameter: 0.01
- Training File: A2--ID3-Training set.csv
- Test File: A2--ID3-TEST set.csv

Who, What, When, Where, Why, How

- Who: DFCA wants a classifier for product order Volume.
- What: ID3 creates a decision tree and IF-THEN rules.
- When: The tree is built from the ID3 training set and tested against the ID3 test set.
- Where: The logic runs in the Python pipeline; each calculation table is exported to Excel.
- Why: ID3 gives readable rules, which helps explain why a Volume class was predicted.
- How: The program calculates entropy, weighted entropy, information gain, pre-pruning alpha, post-pruning loss, and final predictions.
- T1: User-chosen pre-pruning threshold. Higher T1 usually stops more branches early. Lower T1 usually allows deeper splitting.
- g: User-chosen post-pruning parameter, valid range $0 < g \leq 0.015$. Higher g allows more pruning. Lower g allows less pruning.

Formula Checklist

- Entropy/MC = $-\sum(p * \log_2(p))$.
- Gain = $MC(D) - WMC(\text{attribute})$.
- ID3 alpha = $c1 / |F1|$.
- ID3 post-pruning: if $(N - M) / Q \leq gK$, remove subtree.
- Naive Bayes prior = $\text{count}(\text{class}) / \text{total records}$.
- Laplace smoothing = $(\text{count} + 1) / (\text{class count} + \text{number of possible values})$.
- Continuous probability = $1/(\sqrt{2\pi}) * \text{std} * e^{-((\text{value}-\text{mean})^2/(2*\text{std}^2))}$.